ONBRD-07: Understanding Your Data

> **Series:** ONBRD — Dynatrace Onboarding | **Notebook:** 7 of 10 | **Created:** December 2025 | **Last Updated:** 10/02/2026

Exploring What Dynatrace Discovered

With OneAgent deployed, Dynatrace has automatically discovered your infrastructure, processes, and services. This notebook helps you understand what's been found and how to explore your data.

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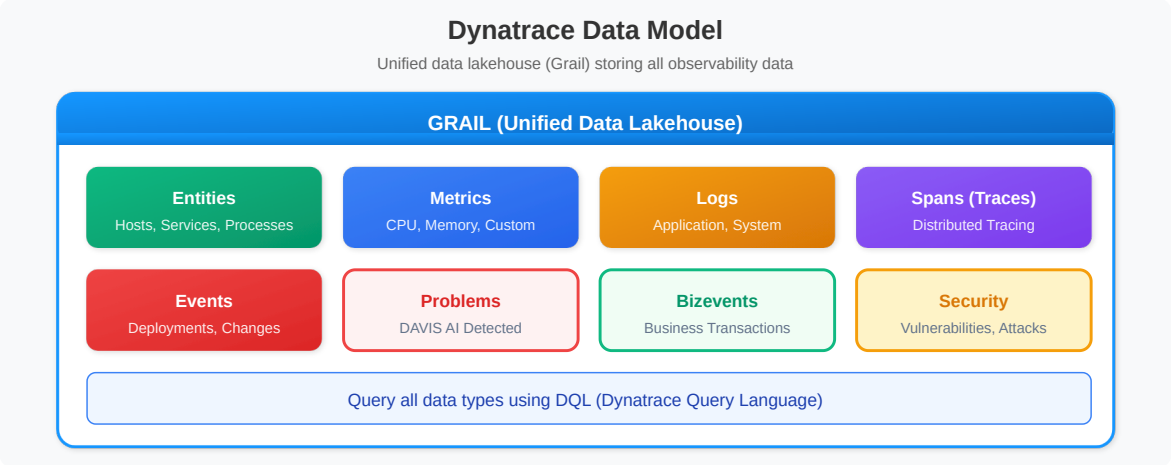
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Prerequisites

- OneAgent deployed on at least one host (ONBRD-05)
- Environment organized with tags (ONBRD-06)
- 15-30 minutes elapsed since deployment for full discovery
- DQL query permissions

1. The Dynatrace Data Model

Dynatrace organizes data into a unified model:

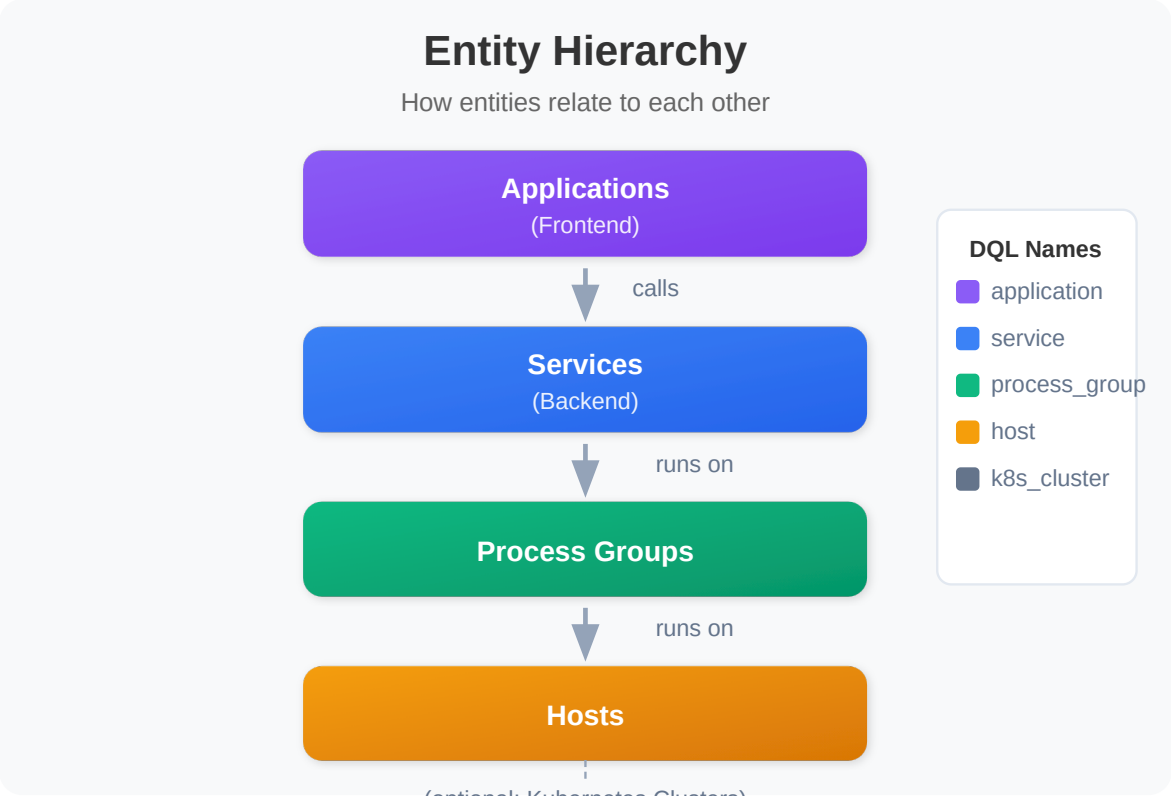


Key Concepts

Concept	Description	Example
Entity	Any monitored component	Host, Service, Process
Metric	Numeric measurement over time	CPU usage, response time
Log	Textual event record	Application log entry
Span	Single operation in a trace	HTTP request, DB query
Event	Point-in-time occurrence	Deployment, config change
Problem	DAVIS-detected issue	Service slowdown

2. Entities and Relationships

Dynatrace automatically discovers and relates entities:



Common Entity Types

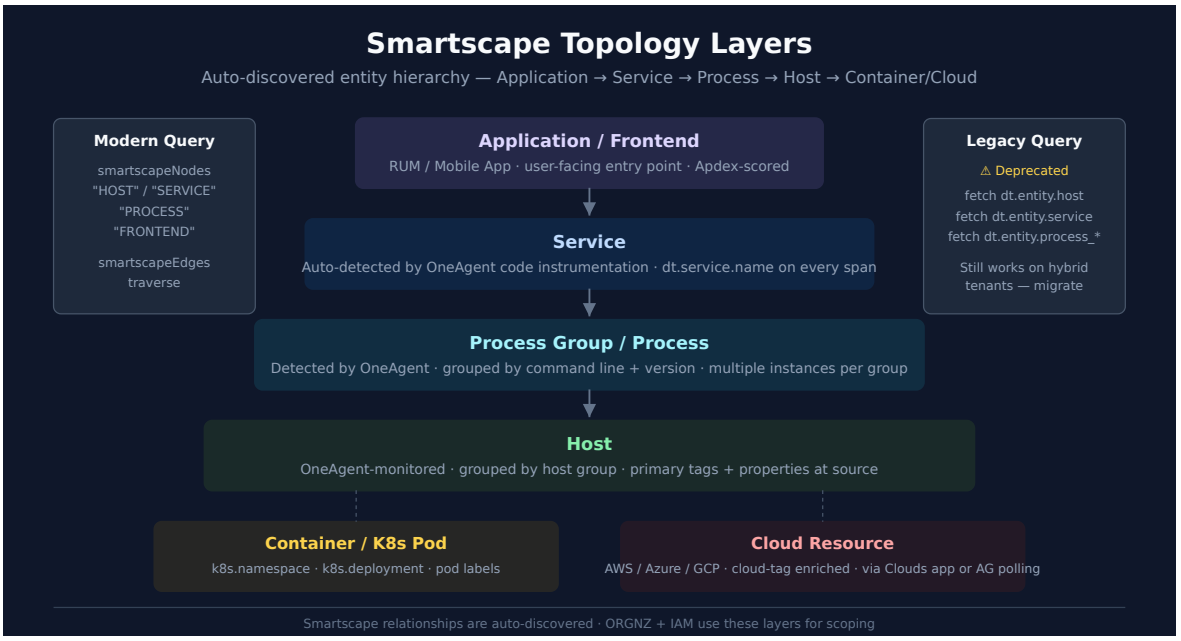
Entity Type	DQL Name	What It Represents
Host	`dt.entity.host`	Physical or virtual machine
Process Group	`dt.entity.process_group`	Set of identical processes
Service	`dt.entity.service`	Logical backend service
Application	`dt.entity.application`	Frontend application
K8s Cluster	`dt.entity.kubernetes_cluster`	Kubernetes cluster
K8s Namespace	`dt.entity.cloud_application_namespace`	K8s namespace
K8s Workload	`dt.entity.cloud_application`	Deployment, DaemonSet, etc.

3. Exploring Topology

The topology view shows the visual representation of your environment.

Location: Infrastructure app → Hosts → Select any host → Dependencies

Understanding Topology Layers



Layer	Shows	Use Case
Applications	Frontend apps, user sessions	User experience
Services	Backend services, APIs	Service dependencies
Processes	Running processes	Process mapping
Hosts	Servers, VMs, containers	Infrastructure view

Reading Topology Views

- **Nodes** = Entities
- **Lines** = Relationships (calls, runs on)
- **Line thickness** = Traffic volume
- **Colors** = Health status (green/yellow/red)

Navigation Tips

- Click any entity to see details
- Use filters to focus on specific services
- Hover over connections to see traffic metrics
- Use the Services app for service-to-service dependencies

4. Data Types in Grail

Grail stores different data types — each queried via a specific DQL command. Each built-in bucket has a documented default retention; custom buckets are configurable.

Data Type	DQL Fetch / Command	Typical Use
Logs	`fetch logs`	Troubleshooting, audit, compliance
Spans	`fetch spans`	Distributed tracing
Metrics	`timeseries` *(not `fetch`)*	Performance monitoring, SLOs
Events	`fetch events`	Change tracking, infra events
Bizevents	`fetch bizevents`	Business transactions, conversion funnels
Davis problems	`fetch dt.davis.problems`	Detected incidents (uses `event.status` / `event.end` fields)
Davis events	`fetch dt.davis.events`	Raw signals that feed problem detection (kind = `DAVIS_EVENT`)
Security events	`fetch security.events`	Vulnerabilities, security signals
RUM sessions	`fetch user.sessions`	RUM sessions (New RUM)
RUM individual events	`fetch user.events`	Page views, clicks, requests, errors
RUM session replays	`fetch user.replays`	Recorded session replays

Data Type	DQL Fetch / Command	Typical Use
Entities (topology)	`smartscapeNodes "<TYPE>"` *(modern)* / `fetch dt.entity.<type>` *(legacy)*	Topology queries; `dt.entity.*` is deprecated, prefer `smartscapeNodes` for new queries

Data Retention

Retention is **bucket-scoped**. Each built-in bucket ships with a documented default retention, and custom buckets can be set from 1 day to 10 years. Inspect your tenant's bucket retention with:

```
// List configured Grail buckets and their retention (use Bucket Management UI for full detail)
fetch dt.system.buckets
| fields name, display_name, dt.system.table, retention_days
| sort name asc
```

Documented defaults for the built-in buckets (confirm against the query above):

Data Type	Built-in Bucket	Default Retention
Logs	`default_logs`	35 days
Spans	`default_spans`	10 days (distributed tracing on Grail is configurable from 10 days to 10 years)
Metrics	`default_metrics`	15 months at 1-minute granularity (Metrics Classic: 5 years)
Events / Bizevents	`default_events` / `default_bizevents`	35 days
Davis problems and events	`default_davis_events`	14 months per the data-retention page; the bucket's own display name reads "Davis events and problems (15 months)" — check yours with the query
Security events	`default_securityevents_builtin` / `default_securityevents`	3 years (Dynatrace-generated) / 1 year (third-party)

Data Type	Built-in Bucket	Default Retention
Self-monitoring and billing events	`dt_system_events`	1 year

> **Sources:** How to organize your data stored in Grail (DT docs) — *"default_logs logs 35 days default_metrics metrics 15 months default_spans spans 10 days"*, Data retention periods (DT docs) — *"Davis problems and events 14 months"*.
>
> **Where to go deeper:**
> - **ORGNZ-02 / ORGNZ-99** — Grail bucket strategy and retention design
> - **OPLOGS series** — Log processing in OpenPipeline
> - **OPMIG series** — Classic Logs → OpenPipeline migration
> - **OPIPE series** — OpenPipeline beyond logs (spans, metrics, events, bizevents)
> - **SPANS series** — Distributed tracing and span analysis

5. Discovery Queries

Use these queries to understand what Dynatrace has discovered in your environment.

Infrastructure Discovery

```
// Count all entity types in your environment. from:-7d also counts entities
that
// have not reported in the last 2 h; without it only the default timeframe
counts.
fetch dt.entity.host, from:-7d | summarize hosts = count()
// Run separately for other types:
// fetch dt.entity.service, from:-7d | summarize services = count()
// fetch dt.entity.process_group, from:-7d | summarize process_groups =
count()

// Smartscape equivalent (dt.entity.* is deprecated but still functional):
//   smartscapeNodes "HOST", from:-7d | summarize hosts = count()
// Caveat: Smartscape can report fewer entities than the classic entity store;
both
// return only entities seen in the query timeframe, so keep the from:-7d.
```

```
// List all hosts with key details
fetch dt.entity.host
| fields entity.name,
         state,
         osType,
         cpuCores,
         physicalMemory
| sort entity.name
| limit 100

// Smartscape note (dt.entity.* is deprecated but still functional): this
query uses the
// classic-only field state, which has NO Smartscape node equivalent
// (Smartscape expresses liveness via node lifetime, not a state field). Keep
the classic
// query above for state detail. For monitoring mode, do not use
monitoringMode — it is
// empty on Kubernetes and Fargate hosts; read billing events instead (ONBRD-
05 § 6).
// Other fields do map: osType -&gt; os.type (LINUX -&gt; OS_TYPE_LINUX);
cpuCores -&gt; cores; physicalMemory -&gt; host.physical.memory; entity.name -
&gt; name.
```

```
// Group hosts by OS type
fetch dt.entity.host
| summarize count = count(), by: {osType}
| sort count desc

// Smartscape equivalent (dt.entity.* is deprecated but still functional):
//   smartscapeNodes "HOST"
//   | summarize count = count(), by: {os.type}
//   | sort count desc
// Caveat: Smartscape can report fewer entities than the classic entity store,
and
// both return only entities seen in the query timeframe (default 2 h) - for a
// discovery inventory add a timeframe such as from:-7d to either query.
// Field maps: osType -&gt; os.type (LINUX -&gt; OS_TYPE_LINUX).
```

Service Discovery

```
// List all discovered services
fetch dt.entity.service
| fields entity.name, serviceType
| sort entity.name
| limit 100

// Smartscape equivalent (dt.entity.* is deprecated but still functional):
//   smartscapeNodes "SERVICE"
//   | fields name, dt.service.sdv1_type
//   | sort name
//   | limit 100
// Caveat: Smartscape can report fewer entities than the classic entity store,
and
// both return only entities seen in the query timeframe (default 2 h) - for a
// discovery inventory add a timeframe such as from:-7d to either query.
// Field maps: serviceType -&gt; dt.service.sdv1_type (SDv1 services only -
null when
// dt.service_detection.version == 2; group by dt.service_detection.version to
see
// which model applies); entity.name -&gt; name.
```

```
// Group services by type
fetch dt.entity.service
| summarize count = count(), by: {serviceType}
| sort count desc

// Smartscape equivalent (dt.entity.* is deprecated but still functional):
//   smartscapeNodes "SERVICE"
//   | summarize count = count(), by: {dt.service.sdv1_type}
//   | sort count desc
// Caveat: Smartscape can report fewer entities than the classic entity store,
and
// both return only entities seen in the query timeframe (default 2 h) - for a
// discovery inventory add a timeframe such as from:-7d to either query.
// Field maps: serviceType -&gt; dt.service.sdv1_type (SDv1 services only -
null when
// dt.service_detection.version == 2; group by dt.service_detection.version to
see
// which model applies).
```

```
// Find services with recent traffic (spans)
// dt.service.name is set on every span, whatever the ingest source
fetch spans, from: now() - 1h
| filter span.kind == "server"
| summarize request_count = count(), by: {dt.service.name}
| sort request_count desc
| limit 20
```

Process Discovery

```
// List process groups
fetch dt.entity.process_group
| fields entity.name
| sort entity.name
| limit 50

// Smartscape note (dt.entity.* is deprecated but still functional):
Smartscape models
// individual processes, not process GROUPS — smartscapeNodes "PROCESS" is a
different
// granularity (process instances), so its results are not comparable to a
process-group
// query. Keep the classic dt.entity.process_group query above.
```

```
// Count process groups
fetch dt.entity.process_group
| summarize count = count()

// Smartscape note (dt.entity.* is deprecated but still functional):
Smartscape models
// individual processes, not process GROUPS — smartscapeNodes "PROCESS" is a
different
// granularity (process instances), so its results are not comparable to a
process-group
// query. Keep the classic dt.entity.process_group query above.
```

Log Discovery

Coverage: what was discovered vs. ingested

> **Coverage vs. volume — the log-module self-monitoring events.** During onboarding, confirm not just *what volume* is arriving but *what OneAgent discovered and whether each source is actually being ingested*. The log module emits a `log_source.status` event per discovered source into `dt.system.events`, carrying `log.source.file_status` and `log.source.ingest_status` — so it reveals sources OneAgent **detected but is not ingesting**, which a `fetch logs` count cannot show (no stored records means nothing to count). It scans events, not raw logs; reading them requires `storage:system:read` on `dt.system.events`.
>
> The log module emits these self-monitoring (SFM) events **by default** — the feature is generally available from **OneAgent 1.339+ and SaaS 1.340+**, and is adjusted through the `builtin:logmonitoring.log-sfm-settings` settings schema. The same events feed the ready-made *Log ingest overview* dashboard. Hosts on older agents send none, so confirm the stream is flowing before relying on it. See **FAQ-08** (Recommended Approach) for the full file-status × ingest-status coverage matrix and the `fetch logs` fallback.
>
> **Sources:** Monitor log source health with SFM events (DT docs) — *"generally available and turned on by default in OneAgent version 1.339+ and SaaS version 1.340+"*, Assign permissions in Grail (DT docs).

```
// Log-source coverage from the OneAgent log-module self-monitoring events.
// Unlike `fetch logs` (which sees only sources that produced records), this
// surfaces sources OneAgent DETECTED but is NOT ingesting. Scans events, not
raw logs.
fetch dt.system.events, from: now() - 24h
| filter event.provider == "Log Module" and event.type == "log_source.status"
| dedup event.id, sort:{ timestamp desc }
| filter event.status == "Active"
| fieldsAdd coverage =
    if(log.source.file_status == "FILE_STATUS_OK" and log.source.ingest_status
== "Ingested", "Available | Ingesting", else:
    if(log.source.file_status == "FILE_STATUS_OK" and log.source.ingest_status
== "Not ingested", "Available | NOT ingesting", else:
    if(log.source.ingest_status == "Partially ingested", "Partial ingestion",
    else: "Unsupported or issue")))
| summarize sources = count(), by:{coverage}
| sort sources desc
```

```
// Check log volume by source
fetch logs, from: now() - 1h
| summarize log_count = count(), by: {log.source}
| sort log_count desc
| limit 20
```

```
// Check log volume by severity
fetch logs, from: now() - 1h
| summarize log_count = count(), by: {loglevel}
| sort log_count desc
```

```
// Sample recent logs
fetch logs, from: now() - 15m
| fields timestamp, loglevel, log.source, content
| sort timestamp desc
| limit 25
```

Kubernetes Discovery (if applicable)

```
// List Kubernetes clusters
fetch dt.entity.kubernetes_cluster
| fields entity.name
| sort entity.name

// Smartscape equivalent (dt.entity.* is deprecated but still functional):
//   smartscapeNodes "K8S_CLUSTER"
//   | fields name
//   | sort name
// Caveat: Smartscape can report fewer entities than the classic entity store,
and
// both return only entities seen in the query timeframe (default 2 h) - for a
// discovery inventory add a timeframe such as from:-7d to either query.
// Field maps: entity.name -&gt; name.
```

```
// List namespaces
fetch dt.entity.cloud_application_namespace
| fields entity.name
| sort entity.name
| limit 50

// Smartscape equivalent (dt.entity.* is deprecated but still functional):
//   smartscapeNodes "K8S_NAMESPACE"
//   | fields name
//   | sort name
//   | limit 50
// Caveat: Smartscape can report fewer entities than the classic entity store,
and
// both return only entities seen in the query timeframe (default 2 h) - for a
// discovery inventory add a timeframe such as from:-7d to either query.
// Field maps: entity.name -&gt; name.
```

```
// List workloads (deployments, etc.)
fetch dt.entity.cloud_application
| fields entity.name
| sort entity.name
| limit 50

// Smartscape equivalent (dt.entity.* is deprecated but still functional):
//   smartscapeNodes {"K8S_DEPLOYMENT", "K8S_DAEMONSET", "K8S_STATEFULSET",
"K8S_CRONJOB"}
//   | fields name, type
//   | sort name
//   | limit 50
// (classic cloud_application spans all of these kinds; add "K8S_JOB" to
include Jobs)
// Caveat: Smartscape can report fewer entities than the classic entity store,
and
// both return only entities seen in the query timeframe (default 2 h) - for a
// discovery inventory add a timeframe such as from:-7d to either query.
// Field maps: entity.name -&gt; name.
```

Problems Discovery

```
// Check for recent problems
fetch dt.davis.problems, from: now() - 7d
| fields timestamp, display_id, event.name, event.status,
affected_entity_types
| sort timestamp desc
| limit 20
```

```
// Problem summary by status
fetch dt.davis.problems, from: now() - 30d
| summarize problem_count = count(), by: {event.status}
| sort problem_count desc
```

6. Next Steps

Now that you understand your data:

1. **ONBRD-08: Your First Queries** — Learn DQL fundamentals
2. Explore topology for dependency visualization
3. Review any detected problems

4. Plan which additional hosts to instrument

Where to Go Deeper

- **ORGNZ-02 / ORGNZ-99** — Grail bucket strategy
- **OPLOGS / OPMIG / OPIPE** — Log and OpenPipeline depth
- **SPANS series** — Distributed tracing depth
- **AIOPS series** — Davis problems, RCA, anomaly detection deep dives
- **K8S series** — Kubernetes monitoring depth

Discovery Checklist

- [ ] Hosts discovered and showing data
- [ ] Services detected and mapped
- [ ] Process groups visible
- [ ] Topology showing relationships
- [ ] Log ingestion working (if applicable)
- [ ] Kubernetes entities visible (if applicable)
- [ ] Bucket retention reviewed against expected data class

Summary

In this notebook, you learned:

- The Dynatrace data model (entities, metrics, logs, spans, events, bizevents, security, RUM)
- How entities relate to each other
- How to navigate topology views
- Different data types stored in Grail and their DQL fetch commands
- That `dt.davis.problems` uses `event.status` / `event.end` fields (not `status` / `end_time`)
- Discovery queries for infrastructure, services, and logs
- That each built-in bucket has a documented default retention, and how to check yours

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References

- Smartscape on Grail (DT docs)
- Service entity in Smartscape (DT docs)
- Grail Data Model
- Smartscape core entities (DT docs)
- Davis Problems App
- How to organize your data stored in Grail (DT docs)
- Data retention periods (DT docs)
- Monitor log source health with SFM events (DT docs)

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.